DGMD E-28
Developing Single Page Web Applications
Assignment 2b: Tic Tac Toe part 2 plus JS Practice

Summary

Your assignment is to continue the Tic Tac Toe game started last week.

Ground Rules

- You may use HTML, CSS, and Javascript
- DO NOT look up how to do this online (you may look up conceptual information)
- You must do each part as specified.
- Use the techniques we reviewed in class and in the notes.
- You may host anywhere you want, but it must be online.
- You may add enhancements or variations after meeting the requirements.

Tasks for this week:

Version 2: ttt_v2.html

Copy ttt_v1 to ttt_v2.

Remove the HTML code to create the nine 's that make up the board and use the JS script below in its place. You may need to adjust your CSS. (*NOTE - you may not change this code*)

```
const NUM_SQUARES = 9;
for (i=0; i<NUM_SQUARES; i++)
{
    id = "sq"+i;
    document.write( "<div class='square' id='" + id + "'></div>" );
}
```

- Ensure that the section to indicate whether it is X's turn or O's turn is a div and put a border around it

- Use a second Javascript script to use Javascript to place 2 X's and one O in the same position you had them in ttt1 (hint: use the innerHTML property)

JavaScript Practice Problems

Add these two problems to a file called practice.html

Problem 1: Create a function called "timesTable" that has one argument called number.
The function should display on the page the times table for that number from 1 to 12.
For example: timesTable(4) would display:

4 x 1 = 4

4 x 2 = 8

4 x 3 = 12

4 x 4 = 16

4 x 5 = 20

4 x 6 = 24

4 x 7 = 28

4 x 8 = 32

4 x 9 = 36

4 x 10 = 40

4 x 11 = 44

4 x 12 = 48

Get a number using prompt() and call the function using that number.

Problem 2: Create a function called fahr2cels() with one argument called ftemp. The function should return the corresponding temperature in Celsius.

You may look up the conversion formulas online.

Here is one that may be helpful: https://www.uptodate.com/contents/calculator-temperature-unit-conversions#:~:text=Temperature%20conversions%20use%20the%20following,*%209%2F5)%20%2B%2032

Call the function with a few different values and display the result using document.write.

Rubric

20 points	Creativity / Effort / Deliverables
50 points	ttt_v2.html meets requirements and works
30 points	practice.html meets requirements and works

Deliverable Worksheet:

URL for ttt_v2.html working online https://konger232.github.io/SAP/03/ttt_v2.html

URL for practice.html working online https://konger232.github.io/SAP/03/practice.html

[✓] All HTML and CSS files have been uploaded to canvas

Answer this question:

Careful planning of the structure of the HTML can assist with identifying selectors in both CSS and Javascript. How do you think the structure of the ttt files so far do or do not help with that goal?

In version 2, the ID of each tile is dynamically generated via JavaScript, meaning one must carefully review the logic to identify specific selector names. Initially, this made identifying selectors slightly more difficult without inspecting the rendered HTML to confirm the mapping. However, the structural advantage of this approach is significant: by generating the div tags dynamically, we can adjust the grid size or tile count entirely within the script. This separates the data from the structure, making long-term maintenance much more efficient than manually editing HTML tags.